

Cyber Threat Intelligence Platform

1. Project Overview

The AI Threat Intelligence & Red Team Platform is an intelligent cybersecurity assistant designed to support both Threat Intelligence (CTI) analysts and Red Team operators. The platform combines MITRE ATT&CK knowledge, threat actor information, detection strategies, and real-world attack procedures using a Retrieval-Augmented Generation (RAG) architecture.

The system leverages:

- MITRE ATT&CK CTI data
- ChromaDB vector databases
- Sentence Transformers embeddings
- Groq-hosted Large Language Models
- Multi-agent architecture

The platform allows users to query cybersecurity knowledge in natural language and receive accurate, context-aware responses.

2. Problem Statement

Cybersecurity professionals face several challenges:

2.1 Fragmented Threat Intelligence

Threat information is distributed across multiple sources:

- MITRE ATT&CK techniques
- Threat actor reports
- Malware databases
- Detection rules
- Security recommendations

Analysts spend significant time searching through these resources manually.

2.2 Difficulty Understanding Attack Chains

Security teams often know individual attack techniques but struggle to understand:

- How attacks are executed in practice
- Relationships between techniques
- Complete attack kill chains
- Real-world attacker procedures

2.3 Slow Incident Investigation

When investigating incidents, analysts need quick answers to questions such as:

- Which threat actor uses this technique?
- How can this attack be detected?
- What mitigation should be applied?
- Which tools are associated with this activity?

Finding these answers manually is time-consuming.

2.4 LLM Hallucinations

Traditional AI assistants may generate inaccurate cybersecurity information because they rely solely on their internal knowledge rather than verified threat intelligence sources.

3. Proposed Solution

The proposed solution is a RAG-based Cyber Threat Intelligence Platform that retrieves validated information from specialized cybersecurity databases before generating responses.

The system:

- Collects MITRE ATT&CK and CTI data
- Converts documents into vector embeddings
- Stores embeddings in ChromaDB
- Performs semantic search on user queries
- Retrieves the most relevant cybersecurity knowledge
- Uses an LLM to generate accurate responses grounded in retrieved data

This approach significantly reduces hallucinations while improving answer quality and explainability.

4. System Architecture

Main Components

Data Layer

The platform ingests cybersecurity datasets including:

- MITRE ATT&CK Techniques
- Threat Actors
- Malware Information
- Security Tools
- Detection Rules
- Mitigations

Embedding Layer

The system converts textual cybersecurity knowledge into vector representations using:

- Sentence Transformers
- BAAI/bge-base-en-v1.5

Vector Database Layer

Knowledge is stored in ChromaDB collections for fast semantic retrieval.

LLM Layer

A Groq-hosted LLM generates final responses using retrieved context.

Agent Layer

Specialized AI agents handle different cybersecurity tasks.

5. Multi-Agent Design

The project contains several specialized agents:

CTI Agent

Responsible for:

- Technique lookup
- ATT&CK framework navigation
- Threat intelligence analysis

Attribution Agent

Responsible for:

- Threat actor identification
- TTP matching
- Campaign correlation
- Attribution scoring

Detection Agent

Responsible for:

- Detection recommendations
- SIEM rule guidance
- Mitigation mapping
- Response actions

Red Team Agent

Responsible for:

- Attack simulation knowledge
- Procedure lookup
- Tool mapping
- Kill chain generation

These agents provide domain-specific expertise while sharing the same knowledge base.

6. Knowledge Base Structure

The platform uses four specialized vector databases:

Database	Purpose
attack_db	MITRE ATT&CK techniques
redteam_db	Attack procedures and sub-techniques
actor_db	Threat actors, malware, and tools
detection_db	Detection analytics, mitigations, and response guidance

This separation improves retrieval accuracy and scalability.

7. Workflow

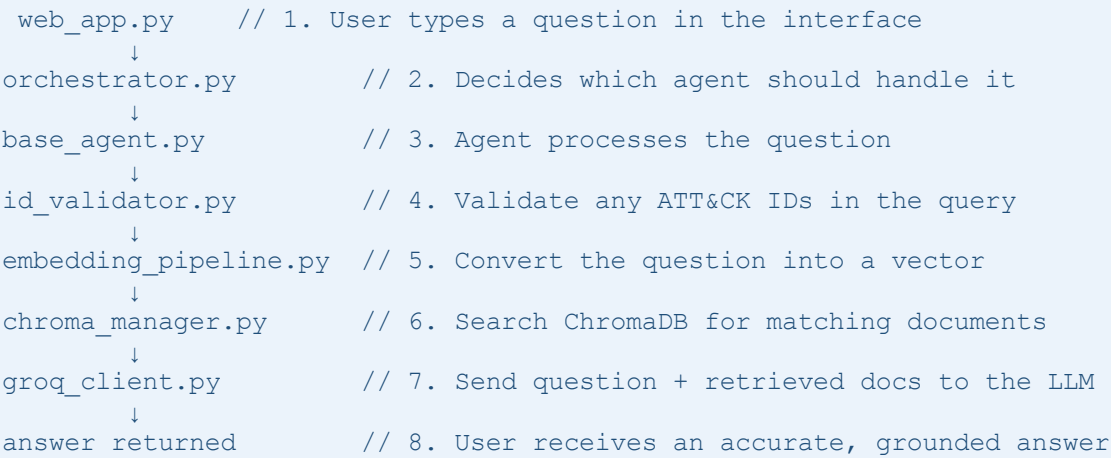
7.1 Ingestion Flow

How data gets downloaded, parsed, and stored before any question is asked.

```
download_mitre.py      // 1. Download MITRE ATT&CK data from the official repo
    ↓
stix_parser.py         // 2. Parse: extract techniques, actors, malware,
tools
    ↓
embedding_pipeline.py  // 3. Convert text into vector numbers
    ↓
chroma_manager.py      // 4. Save vectors into ChromaDB
    ↓
storage/chroma_db/     // 5. Data is ready for search
```

7.2 Question Answering Flow

What happens when a user asks a cybersecurity question.



Example :

Cyber Threat Intelligence Platform

Multi-Agent Cybersecurity Intelligence Assistant

 Chat Assistant  Semantic Search

Ask a cybersecurity question...



 What are the top ATT&CK techniques observed in recent healthcare attacks?

 Agent: CTI Analyst Agent

Based on the retrieved context, I do not see any information about recent healthcare attacks. However, I can provide a general analysis of the techniques mentioned in the context.

The techniques mentioned in the context are:

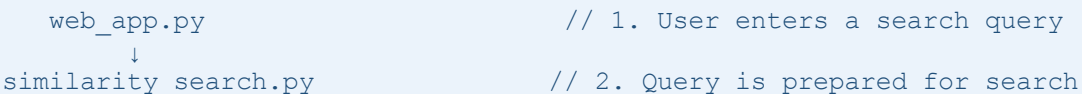
1. T1659: Content Injection
2. T1589: Gather Victim Identity Information
3. T1667: Email Bombing
4. T1557: Adversary-in-the-Middle
5. T1584.006: Web Services
6. T1683.002: Audio-Visual Content

These techniques are not specifically mentioned as being observed in healthcare attacks. However, some of these techniques could potentially be used in healthcare attacks, such as:

- T1659: Content Injection could be used to inject malicious content into healthcare systems, potentially compromising patient data or disrupting healthcare operations.
- T1589: Gather Victim Identity Information could be used to gather sensitive information about healthcare employees or patients, potentially leading to identity theft or other malicious activities.
- T1557: Adversary-in-the-Middle could be used to intercept or manipulate healthcare data, potentially compromising patient care or confidentiality.
- T1584.006: Web Services could be used to compromise access to healthcare-related web services, potentially allowing adversaries to access sensitive healthcare data.

7.3 Semantic Search Flow

A fast, LLM-free mode that returns raw MITRE records directly.



```
↓
embedding_pipeline.py           // 3. Convert query to a vector
↓
chroma_manager.py              // 4. Find similar vectors in ChromaDB
↓
results returned               // 5. Top MITRE records + scores shown
```

Chat Assistant Semantic Search

Semantic Search (No LLM)

Search directly inside the MITRE knowledge bases using vector similarity.

Enter a query

What are the top ATT&CK techniques observed in recent healthcare attacks?

Top Results

6

Search Knowledge Base

6 results found

1. Content Injection

Collection: attack_db

Distance: 0.4179

Metadata:

```
{
  "platforms": "Linux, macOS, Windows"
  "tactics": "initial-access, command-and-control"
  "stix_id": "attack-pattern---43c9bc06-715b-42db-972f-52d25c09a20c"
  "name": "Content Injection"
  "data_sources": ""
  "attack_id": "T1659"
  "is_subtechnique": false
  "type": "technique"
}
```

This mode is fast, deterministic, and useful for debugging retrieval quality or finding relevant ATT&CK content without generating explanations.

8 - Model Comparison: Qwen 3 32B vs Llama 3.1 8B

1. Qwen 3 32B



what are the top attacks technique observed in recent healthcare attacks

Based on the retrieved ATT&CK techniques, the following are the top observed attack techniques relevant to healthcare environments, with their associated risks and healthcare-specific implications:

1. T1566: Phishing

Tactic: Initial Access

Platforms: Windows, macOS, Linux, SaaS, Office Suite

Healthcare Impact:

- **High risk** for healthcare workers via spearphishing emails posing as IT support, vendor communications, or patient-related alerts.
- **Example:** Attackers may embed malicious links or attachments in emails mimicking EHR system updates, leading to credential theft or ransomware deployment (e.g., disrupting patient care systems).
- **HIPAA Relevance:** Unauthorized access to protected health information (PHI) via compromised accounts.

2. T1204: User Execution

Tactic: Execution

Platforms: Windows, Linux, macOS, Containers

Healthcare Impact:

- **Critical risk** for medical device networks and hospital workstations. Attackers may trick users into executing malware (e.g., via phishing-laced

Retrieved Sources

> User Execution

> Gather Victim Identity Information

> Web Services

> Phishing

● Advantages

- Very strong reasoning (better for complex tasks)
- Better coding ability
- Higher quality long answers
- Good at multilingual tasks

➤ Limitations

- More “accurate” in technical explanations
 - Heavier model (needs more GPU/compute)
 - Slower inference
 - Harder to run locally
 - More expensive in production
- ❖ Best for advanced cybersecurity analysis and complex reasoning, but requires more resources and runs slower.

2. Llama 3.1 8B

 what are the attacks technique observed in recent healthcare attacks

 Agent: CTI Analyst Agent

Based on the retrieved context, the following attack techniques have been observed in recent healthcare attacks:

1. **Gather Victim Identity Information (T1589)**: This technique has been used to gather information about healthcare employees' identities, which can be used for targeting. This can lead to **patient safety risks** and **EHR availability/integrity risks**.
2. **Search Threat Vendor Data (T1681)**: This technique has been used by threat actors to gather information about their own campaigns and those conducted by other adversaries, which can help them plan future operations. This can lead to **HIPAA-relevant data exposure risks**.
3. **User Execution (T1204)**: This technique has been used to trick healthcare employees into executing malicious code, which can lead to **patient safety risks** and **EHR availability/integrity risks**.
4. **Email Bombing (T1667)**: This technique has been used to flood healthcare employees' inboxes with spam emails, which can disrupt business operations and lead to **HIPAA-relevant data exposure risks**.
5. **Web Services (T1584.006)**: This technique has been used to compromise access to third-party web services, which can be used for Command and Control, Exfiltration Over Web Service, or Phishing. This can lead to **HIPAA-relevant data exposure risks**.
6. **Social Engineering (T1684)**: This technique has been used to trick healthcare employees into taking actions that result in unauthorized access, approval of changes, disclosure of sensitive information, or execution of adversary-supplied instructions. This can lead to **patient safety risks** and **EHR availability/integrity risks**.

Applicable mitigations include:

- Implementing robust identity and access management controls to prevent unauthorized access to sensitive information.
- Conducting regular threat intelligence briefings to stay informed about emerging threats and tactics.
- Educating employees on social engineering tactics and phishing techniques to prevent user execution.
- Implementing email filtering and spam detection tools to prevent email bombing.
- Regularly monitoring web services for suspicious activity and implementing secure access controls.
- Conducting regular security awareness training for employees to prevent social engineering attacks.

➤ Advantages

- Lightweight and fast
- Easy to run locally
- Lower cost for deployment
- Good general-purpose chatbot
- Efficient for simple tasks

● Limitations

- Weaker reasoning compared to **Qwen 3 32B**
- Less accurate in complex coding/debugging
- Struggles with very long or technical contexts
- Lower quality in advanced analysis

➤ Fast and lightweight for everyday tasks, but less effective for deep technical analysis and reasoning.

★ **Qwen 3 32B** is better for advanced threat intelligence and technical analysis, while **Llama 3.1 8B** is better for fast, lightweight, and cost-effective deployment.

8. Technologies Used

ChromaDB

A vector database used to store embeddings and perform semantic search. It finds the most relevant MITRE ATT&CK records based on the meaning of a user's question.

Sentence Transformers

A machine learning library used to convert text into numerical vectors (embeddings). These embeddings allow the system to compare the similarity between a user's query and MITRE ATT&CK records.

Groq

The Groq SDK allows the platform to communicate with Large Language Models (LLMs). It is used to generate clear and human-readable answers from the retrieved cybersecurity information.

Streamlit

A Python framework used to create the web interface of the platform. It provides an interactive chat application where users can ask questions and receive responses.

Python-Dotenv

A library used to load environment variables from a .env file. It helps manage sensitive information such as API keys and configuration settings securely.

9. Conclusion

The AI Threat Intelligence & Red Team Platform provides an intelligent cybersecurity knowledge assistant built on a RAG architecture. By integrating MITRE ATT&CK data, threat actor intelligence, detection knowledge, and red team procedures into dedicated vector databases, the platform delivers accurate, explainable, and context-aware cybersecurity insights.

The multi-agent design enables specialized analysis for threat intelligence, attribution, detection engineering, and red team operations, making the solution valuable for SOC analysts, CTI teams, and security professionals.